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Radial Artery Pulse Wave Simulator using a Linear Motor

Hyun Heo, Eun Guen Kim, Ki Chang Nam, and Young Huh

Abstract—**To reproduce radial artery pulse wave, electronic simulator was developed. The microprocessor regenerates arterial pulse waveform based on the measured human arterial pulse data. The radial artery pulse wave could be generated from magnetic force using a linear motor. It was also confirmed that developed pulse simulator generated adaptive amplitude depend on the applied pressure on the surface of actuator. It is also necessary to update and extend pulse wave database for many cases and symptoms.**

I. Introduction

THE arterial pulse is the most fundamental sign in clinical medicine. Since 1860, various pulse wave measurement techniques have been developed [1], [2]. In oriental medicine, pulse wave at region of radial artery is one of the important diagnostic parameter of the oriental medical doctors. However, there are few scientific evidences that support the fact of diagnosis. The oriental medical doctor finds pulse position in patient's radial artery by sense of his/her fingers. With palpation at the pulse position, he/she catches pulse wave and diagnoses patient's condition refer to his/her experience. To overcome this scientific weak point, pulse wave measuring system for oriental medicine has been developed based on tonometric method [3]. The major difference between conventional tonometry and pulse wave measurement in oriental medicine is monitoring pulse wave with various applied pressure on the measurement point.

With all the progress of the oriental medicine device, objectification and quantification is still big issue in oriental medicine. And most of oriental medical doctors are required to spend long time to be accustomed to sensing skill and to build their diagnostic experience. Therefore, pulse wave simulator will be helpful for the oriental medical doctor and evaluation of the pulse wave measuring device. Most of researches on pulse wave were focused on fluid dynamics and mathematical modeling of circulation system [4], [5]. Ivison and Wheaton reported about artery pulse wave simulator in 1967, but their simulator was electric signal generator [6]. Some Chinese researchers are developing pulse wave simulator using fluid circulation. Even though their method is close to mimic pulse wave due to blood circulation, it is hard to control arterial pulse wave precisely and the kind of generation pulse wave is limited. In this study, we developed electric pulse simulator to generate quantitative pulse wave through actuator.

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II. Method

A. Pulse wave in oriental medicine

Figure 1 shows how the oriental medical doctor diagnoses a patient's pulse wave. By pressing and raising finger at pulse point, doctor feels pulse characteristics and links it to diseases. In Korea Traditional Medicine, 28 pulse wave formations are used for diagnosis as in Traditional Chinese Medicine [7]. The main parameters to discriminate wave patterns are; pulse width and amplitude, pulse peak duration, artery elasticity and stiffness, pulse rate variability, pulse depth, and etc. Among them, pulse depth means related to artery depth from the skin and it can be evaluated applied force. Undoubtedly, oriental medical doctors don't rely on pulse wave diagnosis entirely, because many conditions such as sex, age, body constitution and etc can affects pulse wave.

Fig. 1. Pulse wave sensing by oriental medical doctor

B. Configuration of pulse simulator

Pulse wave simulator was designed to reproduce human pulse wave by pulsatile vibration using a linear motor. Figure 2 shows the block diagram of the developed pulse wave simulator. External ROM contains database of 150 human radial artery pulse waves. Using the control keypad, user can

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adjust parameters of pulse wave shape and pattern by manual or automatic. And micro processor reproduce pulse wave data based on database and input parameters. Therefore, user can design pulse wave shape in arbitrary.

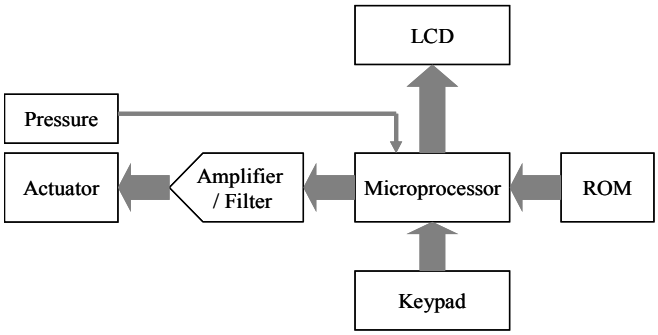

Fig. 2. Block diagram of pulse wave simulator

Figure 3 shows a cross section of the developed pulse simulator's artificial wrist. It produces adaptive pulse amplitude according to finger pressing force. Pressure sensor detects finger pressure which applied pressure on the radial artery and the micro processor control the amplitude of the output pulse.

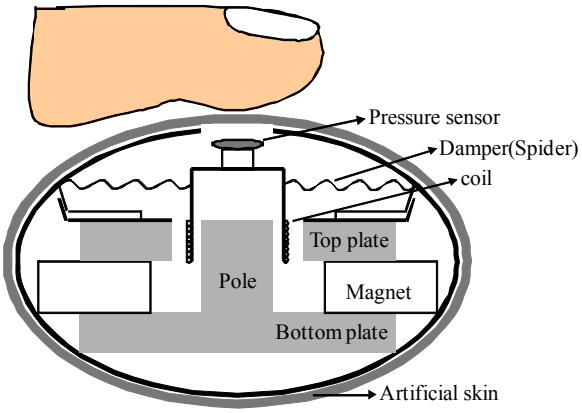

Fig. 3. Cross section of the developed pulse simulator's artificial wrist

Figure 4 is a characteristic of the pressure sensor (IESF-R-5L, CUI Inc., USA). Its active area of diameter is 8 mm and available range of pressure is from 0 to 1.5 kgf (max. 4 kgf). It was also verified that this pressure sensor was tolerable from 0 to 300 mmHg when the pressure was applied using the air cuff.

C. Actuator

To generate arterial pulse, a linear motor was applied. A linear motor operates by responding to positive and negative waves from the amplifier, passing through the coil, causing it to be moved back and forth by attraction and repulsion within the high gauss flux density of the intense magnetic field.

It is widely used in many applications such as computer disk drives, lens focusing, medical equipment, speaker, and etc. Actuator consists of plate, pole, magnet, and coil as shown in Figure 5. It provides the motive force to the spider by the reaction of a magnetic field to the current passing through coil. If the current thought the coil, it induces a magnetic field around the coil. Since the coil is suspended, it is free to move, which is does according to the strengths and directions of the permanent magnetic field and the induced field.

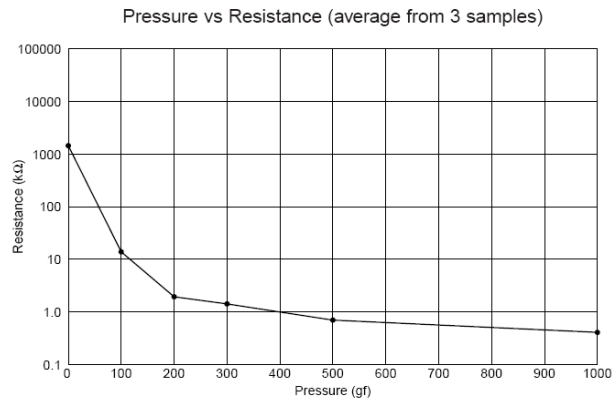

Fig. 4. Characteristics of pressure sensor

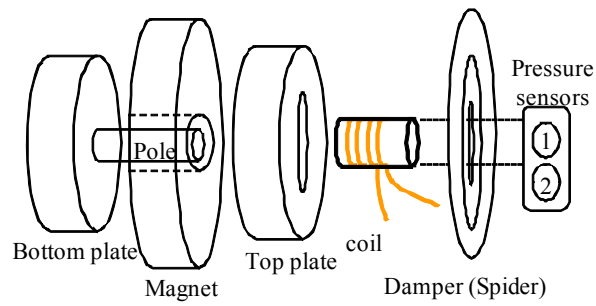

Fig. 5. Voice coil actuator

For pulse wave generation, an actuator is required enough range of motion and be wide frequency response as shown in figure 6. Figure 7 shows the developed simulator, the actuator covers the dynamic characteristic of the arterial pulse.

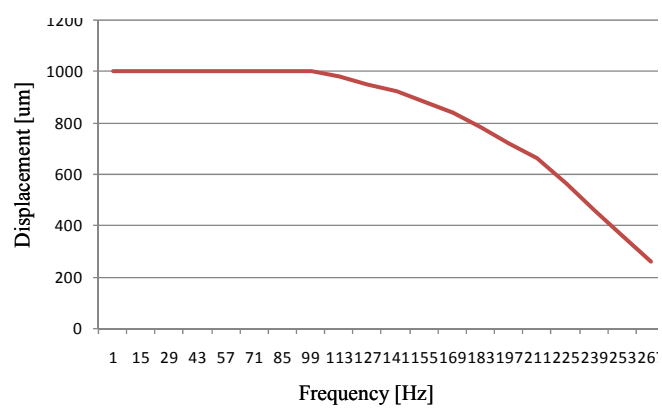

Fig. 6. Frequency response of the actuator

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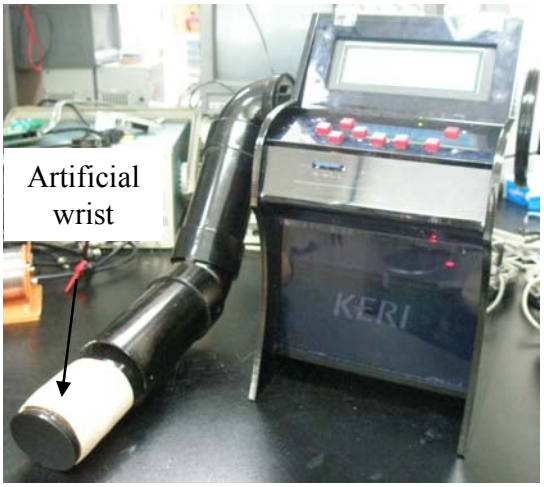

Fig. 7. Developed pulse simulator

D. *Pulse wave DATABASE*

Radial artery pulse waves were measured at left wrist by tonometric method using SphygmoCor PX (AtCor, Austrailia) as shown in figure 8. The measured pulse waves were digitized with 8bit and 128 Hz sampling rate. And digitized data were converted into HEX format to be available in external ROM. Even though the actuator produces the same pulse amplitude as measured by SphygmoCor, the pulse on the surface of pulse simulator might be transferred differently because of structure between actuator and artificial skin. Therefore, digitized data were calibrated with lookup table which was made by comparing with the measured data on the surface of developed pulse simulator.

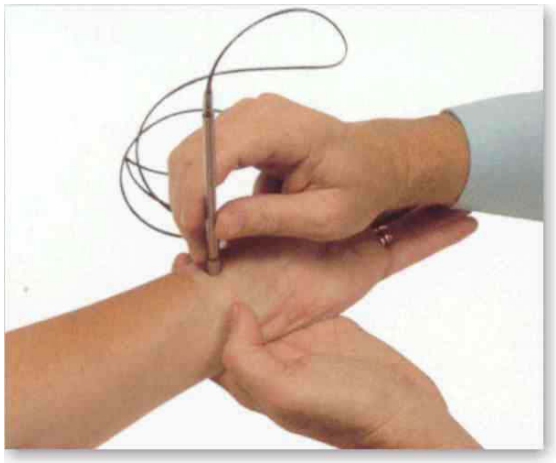

Fig. 8. Pulse wave measurement using SphygmoCor PX (AtCor, Austrailia)

III. Results

Figure 9 shows the measured waves which were generated from the simulator. The artificial wrist contains actuator as shown in figure 8. Pulse simulator was test at various pressures which was applied using air cuff for non-invasive blood pressure monitoring device. Figure 9 shows the measured pulse waves which were generated from pulse simulator according to different applied pressure with 80, 100, and 160 mmHg, respectively. The middle in the figure 9 means unmodified original database wave. Therefore, it generated same amplitude and shape of wave. But the upper and lower waves were modified from the middle wave by sensing the applied pressure. These two waves were measured at different point with distance of 1 cm. Because maximum amplitude of pulse is shown at mean arterial blood pressure (MAP), in case of 100 mmHg which estimated as MAP by preset systolic and diastolic blood pressure produces maximum amplitude.

IV. Discussion

Although fluid dynamic simulator is close to mimic pulse wave transfer because it flow fluid like blood circulation, it is hard to control arterial pulse wave precisely. In terms of pulsatile beating sensing, liner motor is well qualified actuator. Frequency range of arterial pulse is 0.1 – 10 Hz and vibration displacement is less than 2 mm. Therefore, actuator is required to have enough frequency range to cover the pulse wave frequency and to be controlled with subtle up and down movement.

Furthermore, the microprocessor regenerates arterial pulse based on the database according to input parameter such as systolic blood pressure, diastolic blood pressure, pulse peak duration, pulse peak amplitude, applied pressure, and etc. It means that developed pulse wave simulator is free to regenerate various wave formations. But it is necessary to extend clinical database on the various pulse wave and further study on the algorithm to regenerate pulse wave with a combination of input parameters.

Artificial skin also should be taken into consideration, because its mechanical property affects final pulse wave form and applied pressure on artificial wrist.

V. Conclusions

As a result of this study, radial artery pulse wave could be generated using the principle of magnetic force. It was also confirmed that developed pulse simulator generated adaptive amplitude depend on the applied pressure on the surface of actuator. Developed pulse simulator reproduces real human pulse and control the waveform precisely based on the database and input parameters. And it is available for the general pulse simulator at the other body part. Therefore, it is also expecting to be used oriental medical doctor training device and calibration of pulse wave measurement system. Further investigation is necessary about mechanical property of artificial skin and pulse generation to produce the appropriate palpation feeling. It is also necessary to update

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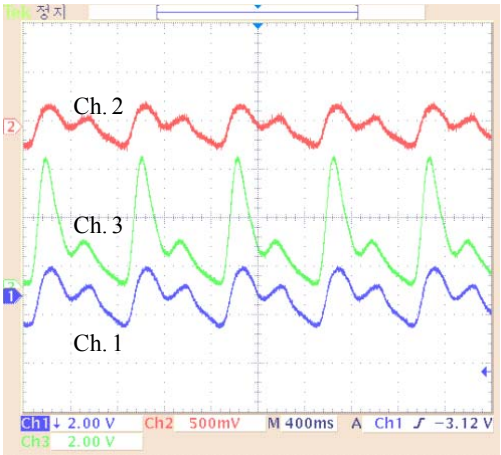
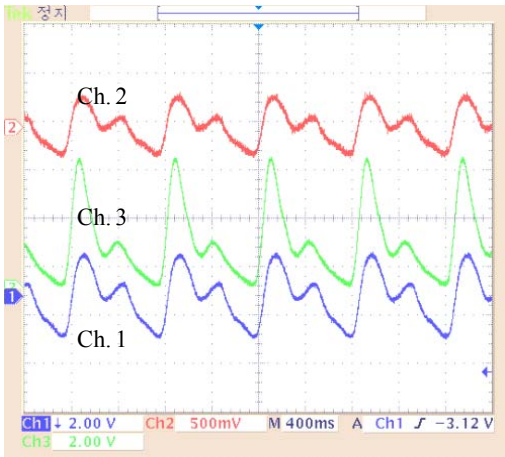
and extend pulse wave database for many cases and symptoms.



(a)



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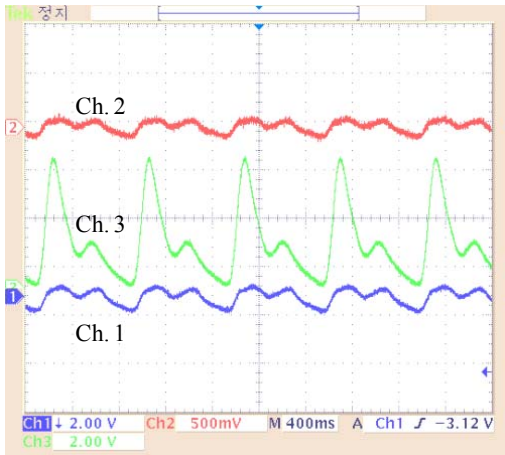

(c)

Fig. 9. Measured pulse waves which were generated from pulse simulator according to different applied pressure (a) 80 mmHg, (b) 100 mmHg, and (c) 160 mmHg, respectively; Ch.1 : measured by sensor (1) in figure 5, Ch.2 : measured by sensor (2) in figure 5, Ch.3 : simulator output from data base

REFERENCES

[1] M. F. O'Rourke, A. Pauca, and X. J. Jiang, "Pulse wave analysis", *Br J Clin Pharmacol*, vol. 51, no. 6, pp. 507-522, June, 2001.

[2] W. W. Nichols and M. F. O'rourke, *McDonald's Blood Flow in Arteries*(5th edition), London, Hodder Arnold, 2005.

[3] H. J. Kang, Y. J. Yoon, "Development of '3-D' Blood Pressure Pulse Diagnosis System by using Intelligent Robot Hand", *13th International Congress of Oriental Medicine*, Oct. 2005.

[4] W. R. Milnor, *Hemodynamics*, Baltimore, Williams & Wilkins, 1989.

[5] V. C. Rideout, Mathematical and Computer Modeling of Physiological Systems, Madison, Medical Physics Publishing.

[6] J. M. Ivison and O. J. Wheaton, "An Arterial Pulse Wave Simulator", *Med. & Biol. Engng*. Vol. 5, pp. 513-518, 1967.

[7] B. Flaws, *The Secret of Chinese Pulse Diagnosis*, Blue Puppy Press, 2006.

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